RTCA, Inc.

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RTCA/DO-160G

Environmental Conditions and Test Procedures for Airborne Equipment

User Guide to Section 21

Emission of Radio Frequency Energy

Based on DO-160G Section 21

Date of Issue:
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Prepared by: RTCA SC-135

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User's Guide, Appendix 21

Section 21 defines limits, test procedures and requirements. In order to properly interpret the procedures and requirements outlined in Section 21, a User's Guide is provided as an Appendix to Section 21.

The User's Guide includes rationale, guidance and background information for the environment, test procedures and requirements, guidance in applying the requirements, and lessons learned from aircraft and laboratory experience. This information should help user's understand the intent behind the requirements, should aid in tailoring requirements as necessary for particular applications and should help user's develop detailed test procedures based on the general test procedures in Section 21.

The User's Guide follows a parallel format to the main body of Section 21 for easy cross reference. Paragraph numbers corresponding to each paragraph in Section 21 are included the User's Guide. Where there is no additional information provided in the User's Guide, the paragraph header is provided but the following paragraph is left blank.

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21.0 Emission of Radio Frequency Energy

21.1 Purpose of the Test

21.2 Equipment Categories

21.3 General Requirements

21.4 Conducted RF Emissions

21.5 Radiated RF Emissions

21.6 Radiated RF Emissions; Reverberation Chamber Procedure

21.6.4 Radiated RF Emissions Test

i. During the development of the reverb technique it was noted that the limit lines originally incorporated into Section 21 were based on the directivity of a dipole. This led the working group to conclude that the assumption in Note 1 was in keeping with the original limit lines.

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